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REVIEW

A multimodal approach to eye melanoma: patterns of care and related complications

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Abstract We describe the results of multimodal treatment of uveal and conjunctival melanomas. A retrospective analysis was performed on 54 patients treated with a multimodal approach between 2003 and 2008 in a single institution. Main outcome measures were survival, enucleation rate, local tumor control, visual function preservation and complications associated with treatments. The median follow-up was 33.4 months. The 5-year overall survival was 95.3%, the local recurrence was 3.7% and the 5-year enucleation was 9.4%. Vision preservation was achieved in 84% of cases. Observed complications were cataract, retinal detachment, diplopia, glaucoma, retinopathy, optic neuropathy and scleral necrosis. A careful consideration of treatment of uveal melanoma in this study allowed us to obtain the survival rates and visual outcomes similar to previously published results, with a very small incidence of complications. the results must be interpreted in the light of recent findings on the genetic pattern of uveal melanoma.

Keywords Melanoma · Uvea · Conjunctiva · Eye · Brachytherapy · Plaque · Bevacizumab

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Introduction

Uveal melanoma is an uncommon pathology occurring mainly in older adult patients without a clear hereditary nature. Generally, it is caused by an accumulation of multiple genetic lesions and is fatal in a relevant fraction of cases, even if a successful local control is achieved [1, 2]. Several clinical, pathological and cytogenetic features seem to have prognostic value for metastatic disease, such as older patient age, larger tumor size, anterior tumor location, epithelioid histotype and aneuploidy 3 [3–7]. A few recurring chromosomal alterations are associated with the increased risk of metastasis and in particular loss of chromosome 3 and gain of 6p and 8q, but no specific individual genes responsible for carcinogenesis have been clearly demonstrated in these loci [8, 9]. Recent investigations based on gene expression profiling have shown that uveal melanomas cluster naturally into two groups, more specifically low-grade tumors (class 1) and high-grade tumors (class 2), based on gene expression signature [2, 10, 11]. These two molecular classes can be related to chromosome 3 status; in particular, only class 2 tumors display down-regulated gene clusters on chromosome 3 and up-regulated clusters on chromosome 8q. The genic identifications of these two tumor classes can be the key to also understand the biological processes promoting metastatization. Correspondingly, it has been found that greater basal dimension and greater distance from the optic disc are factors associated with monosomy 3; moreover, this aneuploidy has been noted in 26% of small size, 36% of medium size and 33% of large size melanomas [2, 12]. The possibility of determining the patient's specific risk of metastatic spread, not only will influence future research directions, but also the current clinical practice. In view of new therapeutic strategies based on a complete comprehension of molecular mechanisms of this pathology, it should be useful to discuss the current clinical experience, its appreciable issues and controversial aspects. From a clinical point of view, systemic chemotherapy in metastatic disease has not been proven to obtain significant impact on survival, while liver-targeted treatments are under investigation. Radiotherapy is the main determinant of successful conservative strategies against ocular melanoma. In particular, brachytherapy (BT) [13] involves many advantages with respect to other treatment modalities like charged-particle therapy [14–19], stereotactic radiosurgery (SRS) [20, 21] or external beam radiotherapy (EBRT) [22, 23], although it ensures an equivalent therapeutic efficacy. The use of brachytherapy in intra-ocular neoplastic pathologies allows us to conform the dose to the target volume, avoiding geographical miss related to organ motion and sparing healthy tissues; thus, the incidence of enucleation for local disease progression or radiation-related adverse events is very low [24–27]. The specific characteristics of this radiotherapeutic modality has given it an invaluable role in the primary treatment of the uveal malignant melanoma (UMM), a scarcely radio-sensitive pathology. During the two last decades, the multicentric collaborative ocular melanoma study (COMS), promoted by the National Cancer Institute, the National Eye Institute and the National Institute of Health, compared the efficacy of enucleation versus BT with 125-I X-ray emitting sealed sources; the COMS guidelines are currently considered the standard reference in the management of UMM [28–31].

Radioactive seeds can be assembled on gold episcleral plaques in order to adapt the dose surfaces to the particular shape and extension of any individual case [32]; moreover, the emitted radiation is quite completely confined within the patient's eye and the risk associated to radiation exposure for the public and the personnel during treatment is very low. Also, 106-Ru plaques have gained a great popularity, due to their relatively low activity, their specific beta emission, the 1-year half-life, their steepest spatial dose gradient, their ready-to-use characteristics and the wide variety of shapes and dimensions [33–36]. Nowadays, 125-I and 106-Ru plaques have a well-consolidated role in the primary treatment of uveal and conjunctival melanomas; although they are not perfectly interchangeable, the choice of isotope is also based on radiotherapy center tradition and on available devices. SRS and heavy-charged-particle treatments are accessible only in specialized centers and selection of these options also depends on the expertise of the treating physician.

In 2003, our Hospital began to propose an integrated multimodal approach, based on personalized interventions, to ocular neoplastic pathologies, addressing a particular attention to organ sparing and visual function preservation issues. To assess the safety and the effectiveness of multimodal treatments of UMM and planning to integrate diagnosis with genomic analysis, we reviewed the results obtained in a 5-year period, with a retrospective, non-comparative, monocentric study.

Methods and materials

Patient selection

This study aims to retrospectively review clinical data relative to 54 patients affected by uveal or conjunctival melanomas, who received a radiation treatment at our center from May 2003 to July 2008. Clinical data collection included age at treatment time, gender, tumor laterality, localization, and TNM classification [37]. Table 1 reports the internationally agreed classification criteria for uveal and conjunctival melanomas.

Tumor characteristics and therapeutic approach

Patients with diagnosis of uveal or conjunctival melanoma were subject to ophthalmologic examination, including US study of the eyes, fluorescein angiography, indocyanine angiography and CT or MRI scan of the orbital region. Moreover, clinical examination and thorax and abdomen CT scans with contrast medium were required to exclude the presence of metastases. The treatment was then configured on the basis of clinical findings. In our series of patients, BT was the treatment of choice for small- or medium-sized melanomas. A 106-Ru plaques were generally indicated for T1–T2 lesions with a single administration; in case of medium size T2–T3 melanomas, the treatment was performed in a single fraction with a 125-I plaque, or in two subsequent fractions with a 106-Ru and a 125-I plaque, respectively, the latter option was adopted to keep the scleral dose under unacceptable toxicity levels and ensure an optimal dose coverage.

In the presence of large, locally advanced melanomas, an SRS primary treatment was administered. A sequential brachytherapy treatment was performed 6 weeks after SRS to enhance the vascular damage within the tumor [38, 39]. Primary or salvage enucleation was performed in patients with locally advanced tumors. An SRS treatment was also performed with prophylactic intent before enucleation. Whenever possible, surgical resection was indicated as primary treatment of conjunctival melanomas, followed by BT. Criteria for planned volume individuation were based on treatment modality. BT target volume included a safety margin to lateral borders of the tumor base. A margin was also considered over the tumor apex only in the presence of large collar-button shaped melanomas, satellite cysts or exudative detachments. Dose prescription was made at the

Table 1 Internationally agreed classification criteria for uveal and conjunctival melanoma

Type		AJCC/UICC TNM classification for uveal and conjunctival melanoma			
		Choroid	Iris	Ciliary body	Conjunctiva
TX			Primary tumor cannot be assessed		
T0			No evidence of primary tumor		
T1	TIA	Height ≤ 2 mm and base ≤ 7 mm	Limited to iris	Limited to ciliary body	Limited to bulbar conjunctiva
	TIB	Height ≤ 3 mm and base ≤ 10 mm			
T2		Height ≤ 5 mm and base ≤ 15 mm	≤1 Quadrant, invasion of nearest structures	Invasion of iris or anterior chamber	Extension to cornea
T3		Height > 5 mm and base > 15 mm	>1 Quadrant, invasion of nearest structures	Invasion of choroid	Invasion of conjunctival fornix or palpebra
T4			Invasion of nearest organs		
NX			Regional lymph nodes cannot be assessed		
N0			Negative for regional lymph node metastases		
N1			Positive for regional lymph node metastases		
MX			Distant metastasis cannot be assessed		
M0			Negative for distant metastases		
M1			Positive for distant metastases		

COMS classification of uveal melanoma following dimensions

Type	Maximum height (mm)	Maximum base diameter (mm)
Small tumors (S)	<3 and	≤10
Medium tumors (M)	≤10 and	≤16
Large tumors (L)	>10 or	>16

tumor apex or at the COMS prescription point, depending on tumor extension. When SRS was the appropriate treatment, lesion volume with a safety conformation margin was identified on a set of axial CT orbital images and dose was prescribed at a central target point. A trans-pupillary thermo-therapy (TTT) with an infrared diode laser (810 nm) was performed 4–6 months after radiotherapy when a sub-optimal tumor regression was found [40].

Plaques dosimetry

Our center employed both 125-I Isoseed model sources fixed on custom gold episcleral shells and 106-Ru plaques (BEBIG Isotopen und Medizintechnik GmbH, Berlin DE). Basically, the choice of the specific isotope was carried out in relation to the extension of the tumor. In all, 125-I seeds allowed us to encompass the whole target volume within the prescription isodose also in the presence of large lesions and to maintain the dose to the sclera within tolerable values. Moreover, seed distribution and orientation could be tailored for any specific situation, using a standard multipurpose plaque or by preparing a gold shell with custom template. On the other hand, 106-Ru plaques ensured a steeper lateral dose gradient and, therefore; they allowed the best sparing of the nearest structures at risk. In all cases, the tumor base of implant was comprised within the active surface with a lateral margin greater than 2 mm in all directions. All radioactive sources were supplied with a calibration certificate traceable to an international primary standard dosimetric laboratory. All 125-I seeds were also calibrated at our center with a well-type ionization chamber of Standard Imaging, model HDR1000 Plus (Standard Imaging Inc. Middleton – WI), with a traceable calibration in air kerma strength for Isoseed geometry. Moreover, 106-Ru plaques were calibrated independently at our center with an OPTIDOS plastic scintillation dosimeter type

10,013–0003 (Optidos PTW, Freiburg - DE), following the indications of the IAEA dosimetric protocol for low energy X-rays or beta brachytherapy sources [41]. The Optidos dosimeter has a sensitive plastic cylindrical element, 1 mm in diameter and 1 mm in height, connected with a 1 m fiberoptic cable to an electrometer. The dosimeter was supplied with a traceable certificate of calibration in dose to water for 106-Ru emission and it was also calibrated in dose for beta sources with a similar energy spectrum at ENEA Italian National Primary Ionizing Radiation Metrology Laboratory [42].

Treatment planning

Patients undergoing SRS were subject to head immobilization with a thermoplastic mask to a stereotactic frame (3Dline s.r.l., Milano—I); a series of axial CT cranial slices were then acquired. A blinking LED, oriented towards the patient's face, was fixed to the head frame and patients were instructed to fix the light during CT scan and treatment with the treated eye (if sighted), or with the fellow one. Target volume and organs at risk were individuated and contoured on CT images. SRS treatments were planned with Ergo treatment planning system (3Dline s.r.l., Milano—I); a treatment plan was composed of a maximum of six dynamically conformed isocentric arcs. Each arc was divided in segments of 20°–40° with independent dose contributions. A dose-constraint optimization was then performed by applying a built-in algorithm. A 30 Gy dose was prescribed in single fraction, or a 50 Gy dose was administered in five daily fractions. Patients undergoing SRS were irradiated with a 6 MeV Siemens Primus linear accelerator equipped with a 3Dline MicroMultiLeaf collimator. BT planning was performed with a BEBIG Plaque Simulator v4. Fundus photographs, eye ultrasound and CT or MRI data were included in the treatment plan in order to accurately evaluate the doses to the structures at risk for radio-induced effects, i.e. inner sclera, lens, disc and macula. A manual dose calculation was also performed to verify the treatment duration. The prescribed dose was 100–120 Gy to tumor apex, for a lesion less than 5 mm in height and for 106-Ru treatment; for larger lesions treated with 125-I plaques, a 85 Gy dose was prescribed to the COMS point (on the central axis of the plaque, 6 mm away from its concave surface) or to the lesion apex for higher tumors. A minimum distance of 1 mm from the tumor base to the plaque distance was considered to take account of the sclera.

Surgical procedure for brachytherapy

Ophthalmologists applied the plaque under local anesthesia as an outpatient procedure. Conjunctival tumors were outlined with a dermographic pen and then excised. Dealing with intraocular tumors, a 360° peritomy was fashioned and the four rectus muscles were identified and isolated. Using transillumination the tumor was localized, margins were outlined on the globe using dermographic pen. Tumors localized to the posterior pole also required confirmation with indirect ophthalmoscopy and scleral depression. If necessary, muscles overlapping the treatment area were temporarily detached, to ensure a correct positioning of the plaque. A transparent dummy plaque was used to simulate the plaque placement and to position the scleral sutures. The plaque was then fixed to the sclera and its position was verified with US scan. The patient was surveyed for radiation protection purposes before being discharged. Plaque removal was performed under local anesthesia at the scheduled time and the effective treatment duration was recorded.

Follow-up

Survival was defined from the beginning of the treatment. Patients were followed up in combined clinics with radio-oncologists and ophthalmologists at weeks 1, 4, 12 and thereafter every 3 months or in presence of new symptoms. Ultrasound eye scans were performed every 3 months and fluorescein and indocyanine green angiography were performed every 6 months. Local control was established if the mass-lesion decreased on the fundus examination and echographic modifications were appreciable (reduction of more than 1 mm in thickness and or increase in reflectivity of more than 20%) [43]. If the lesion remained unaltered, local control was scored and a close follow-up was performed. In case of local progression, a treatment failure was recorded and the eye was subject to salvage surgical enucleation. In any case, follow-up was continued. When multiple distant metastases were present, enucleation was proposed only in cases of unbearable symptomatology. Only metastatic patients received chemotherapy. Objective of the study and statistical analysis.

Outcome measures of the study were local control (LC), globe sparing (GS), best corrected visual acuity (BCVA) preservation, incidence of complications, overall survival (OS) and disease-free survival (DFS). Vision preservation was defined if either a residual best corrected visual acuity (BCVA) of more than 20/200 or a loss lower than two lines from the baseline value were maintained. Kaplan–Meier event-free survival curves were computed by using the product-limit method [44]. Differences of therapeutic success between subgroups of patients were tested with the Fisher exact test and statistical quantitative analysis of therapeutic success was assessed with the Dunn test. All adverse events related to the treatment were scored following CTCAE criteria [45].

Results

From May 2003 to July 2008, 54 patients were treated at our center for melanoma of the uvea or of the conjunctiva. Table 2 reports the main characteristics of this group of patients, together with a repartition of tumor sizes and localizations. No patient showed nodal involvement; metastases were present in two patients at diagnosis. Median follow-up was 33.4 months. Table 3 reports a detailed analysis of therapeutic approaches. TTT was performed in 10 patients (18.5%) in which a sub-optimal tumor regression was evidenced. EBRT treatment was performed with electrons for a patient with a large conjunctival melanoma extending to the caruncle.

Tumor control

At analysis, 51 patients (94.4%) were still alive. Two patients with evidence of hepatic metastases at diagnosis died for uncontrolled disease progression; one patient in complete remission for UMM died for a major adverse cardiac event (Table 4). Failure of disease control was registered in five patients (9.3%); in two of them (3.7%) only a lesion mass increase (local failure) was evidenced and one patient (1.9%) was subject to local failure and distant metastases. LC rate was 96.3%. Despite LC, metastatic progression occurred in two patients (3.7%). Post-therapy assessments evidenced a median tumor height reduction of 1.2 mm.

Table 2 Characteristics of patients

Pateints characteristcs	
Number of patients	54
Male/female	19 (35.2%)/35 (64.8%)
Median age in years (range)	70.4 (34.7–90.5)
Median follow-up time in months (range)	33.4 (3.9–60.1)
Anatomical localization of the tumor	
Choroid	45 (83.3%)
Iris	2 (3.7%)
Ciliary body	4 (7.4%)
Conjunctiva	3 (5.6%)
Geographical localization of the tumor	
Median longitude/latitude (deg)	172/−5
Right eye/left eye	30 (55.6%)/24 (44.4%)
Tumor dimensions (in mm)	
Median largest diameter (range)	10.5 (5.0–20.2)
Median height (range)	3.35 (1.0–13.0)
COMS classification of uveal melanomas (51 patients)	
Small size	4 (7.8%)
Medium size	37 (72.5%)
Large size	10 (19.7%)
Global TNM classification (54 patients)	
T1A (uveal MM)	2 (3.7%)
T1B (uveal MM)	16 (29.5%)
T1 (conjunctival MM)	1 (1.9%)
T2	15 (27.8%)
T3	19 (35.2%)
T4	1 (1.9%)
M0	53 (98.1%)
M1	1 (1.9%)

Table 3 Analysis of pattern of care and radiotherapy characteristics

Therapeutic choices	
Surgical resection	
Followed by 106-Ru plaque BT	2 (3.7%)
106-Ru BT	24 (44.4%)
Followed by TTT	5 (9.2%)
125-I BT	9 (16.6%)
Followed by TTT	3 (5.5%)
Bi-fractionated BT with 106-Ru and 125-I	1 (1.9%)
Followed by TTT	1 (1.9%)
SRS	
Followed by TTT	1 (1.9%)
Followed by 125-I BT	3 (5.5%)
Followed by 125-I and 106-Ru BT	1 (1.9%)
Followed by salvage enucleation	2 (3.7%)
Neo-adjuvant to enucleation	1 (1.9%)
EBRT (electrons)	1 (1.9%)
Doses in Gy to structures at risk (ranges)	
Macula	6.41 (0–177)
Lens	14.4 (0–72)
Disc	5.1 (0–126)
Inner sclera (tumor base)	293 (50–826)

BT brachytherapy, *EBRT* external beam radiotherapy, *SRS* stereotactic radiosurgery, *TTT* trans-pupillary thermo-therapy

Table 4 Analysis of results

Objective response	
Global disease control	49 (90.6%)
Local progression only	2 (3.7%)
Distant metastases only	2 (3.7%)
Local progression and distant metastases	1 (1.9%)
Tumor changes from basal to follow-up values in mm	
Median base diameter variation (range)	2.11 (0–13.8)
Median height variation (range)	−1.2 (−6.9–1)
Visual outcome	
Lowering of BCVA	13 (24.0%)
Unvaried BCVA	34 (63.0%)
Rise of BCVA	7 (13.0%)

BCVA best corrected visual acuity

Primary or salvage enucleation was performed in three patients (5.6%). Enucleation due to radiotherapy complications (such as symptomatic neovascular glaucoma) was required in another two patients (3.7%). All enucleations were performed during the first 25 months after radiotherapy. Therefore, the 1-year and 5-year enucleation probability were, respectively, 5.7% (95% CI = ±6.2%) and 9.4% (95% CI = ±10.3%) (Fig. 1). The 1-year and 5-year OS were, respectively, 100 and 95.3%. The 1-year and 5-year DFS rate estimated with the Kaplan–Meier curves were, respectively, 100% (95% CI not determined) and 85.6% (95% CI = ±7.9%). It should be noted that failures occurred in three out of ten patients with large UMM (30%), in two out of 37 patients with medium UMM and in the patient with the largest of the three conjunctival melanomas (globally, 7.9%), and in none of the patients with small uveal/conjunctival melanomas. For the 51 patients affected by UMM, disease control was assessed separately for each one of the three groups of patients having, respectively, small, medium and large lesions; their respective 5-year rates of disease control resulted to be 100, 94.6 and 70% [n (small) = 4, n (medium) = 37 and n (large) = 10]. A statistically significant difference between the 5-year DFS rates of, respectively, large UMM class and medium and small UMM classes merged together ($P = 0.049$). Therefore, the relative tumor height reduction with respect to the basal value was analyzed separately for the three groups of patients. It was found that the percent height reductions were, respectively, 14% for large UMM, 40.5% for medium UMM and 66% for small UMM, and they were significantly different ($P = 0.01$).

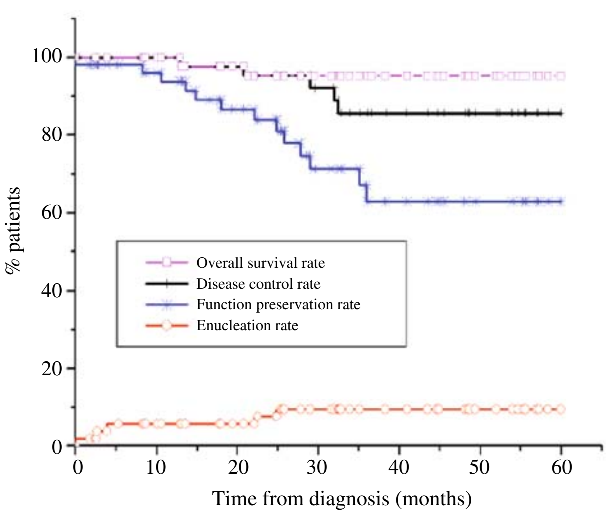

Fig. 1 Kaplan–Meier curves for overall survival (*squares*), disease-free survival (*crosses*), function preservation probability (*stars*) and enucleation rate (*circles*)

Functional outcome

Radiotherapy generally allowed us to preserve a useful BCVA. A visual function impairment was reported in 13 patients (24.0%): 11 of them (20.3%) were subject to a loss greater than two lines and two of them (3.7%) maintained a residual BCVA lower than 20/200. Thirty-four patients (63.0%) maintained their visual function unchanged and seven patients (13.0%) had a rise of BCVA (Fig. 1). Therefore, the ratio of successful functional outcome in patients not enucleated was (34 + 7)/49 = 0.84.

Adverse effects and other therapies

The most frequent acute event registered in the present study was the painful sensation (grade 1–2 in 72% of patients) during brachytherapy treatment. Twenty-five patients manifested one or more late complications due to treatment (Table 5). The major late adverse effects were

Table 5 Analysis of the rate of complications

Ocular/visual adverse effects scoring				
Event	Grade	n (%)	Treated	Resolved / enucleation
Cataract	G2	1 (1.9)	0	1/0
	G3	9 (16.7)	9	9/0
Diplopia	G1	2 (3.7)	0	2/0
	G2	3 (5.5)	3	3/0
Glaucoma	G2	3 (5.6)	3	3/0
	G3	1 (1.9)	1	1/0
	G4	1 (1.9)	1*	0/1*
Retinal detachment	G2	4 (7.5)	4	4/0
	G3	2 (3.7)	2	2/0
Retinopathy	G1	1 (1.9)	1	1/0
	G2	1 (1.9)	1	1/0
	G3	3 (5.6)	3	3/0
Scleral Necrosis	G4	1 (1.9)	1	0/1
Optic nerve atrophy	G2	1 (1.9)	1	1/0
	G3	1 (1.9)	1	1/0
Pain (acute only)	G1	18 (33.3)	0	18/0
	G2	11 (20.4)	11	11/0
Pain (late)	G4	1 (1.9)	1*	0/1*
Treatments for complications				
PE and IOL				9 (16.7 %)
Small prism correction				3 (5.6 %)
Intravitreal bevacizumab				6 (11.1 %)
Cyclocryo-coagulation				1 (1.9 %)
Enucleation				2 (3.7 %)
Argon laser photocoagulation				5 (9.3 %)

The items marked with "*" refer to the same eye

IOL intra-ocular lens, *PE* phako-emulsification cataract ($n = 10$), exudative retinal detachment ($n = 6$), diplopia ($n = 5$), glaucoma ($n = 5$), radiation retinopathy ($n = 5$), optic nerve atrophy ($n = 2$) and scleral necrosis ($n = 1$).

For all patients, doses to structures at risk, estimated by the treatment planning system, were recorded. Nine patients with grade 3 cataract (6 unilateral and 3 bilateral) underwent phako-emulsification with intraocular lens (IOL) implant. Dose to lens for the whole population of patients ranged from 0 to 72 Gy; the two groups of patients, respectively, with and without cataract development received doses to lens which were not statistically different ($P = 0.39$). Intravitreal bevacizumab (Genentech, Inc.; South San Francisco, CA) was administered postoperatively to six patients in order to reduce postoperative exudative retinal detachment. Diplopia resolved spontaneously in two patients (3.7%) and required a small prism correction in three patients (5.6%). Three patients affected by secondary grade 2 glaucoma received topical medical therapy; one patient with grade 3 glaucoma required cyclocryo-coagulation and one patient was enucleated due to uncontrolled grade 4 glaucoma. Mild to moderate radiation retinopathy occurred in five patients and was successfully treated with argon laser photocoagulation and intravitreal bevacizumab. Two patients with optic disk edema and nerve atrophy received a dose to disk of 4.8 and 18 Gy, respectively (population range was 0–96 Gy). One patient with a very large UMM was treated with a 50 Gy fractionated SRS and a 45 Gy 125-I BT treatment, receiving a total scleral dose of 308 Gy. Eight months after treatment, she developed a scleral necrosis and was treated with surgical implant of a scleral patch, but after 2 months the eye had to be enucleated, due to persistence of symptomatology.

Discussion

Management of uveal and conjunctival melanomas has evolved over past decades with a trend toward the combination of multiple focal conservative treatments. The COMS experience [31] indicates that the treatment choice (enucleation vs. conservative approach with 125-I plaques) does not affect patient survival. Photocoagulation by means of xenon or argon ion lasers has been used in the past as an option in selected small choroidal melanomas. Recently, it has been replaced by TTT that has been proven to be effective in treating selected small volume lesions. Currently, TTT is also used as a supplement to plaque radiotherapy [40], to decrease the incidence of intraocular tumor recurrence and severe late effects. Trans-scleral tumor resection has been reported for the treatment of selected iris, ciliary body, or anterior choroidal as an alternative or in combination with plaques. However, a longer follow-up is necessary to establish the efficacy on tumor control [46]. Radiotherapy remains the most used approach. Plaque based BT has been shown to be effective, cheap and extremely flexible as customized plaques can be produced to deal with almost all clinical situations. Unfortunately, many patients treated with radiotherapy for choroidal melanoma experience sight-limiting side effects, and a variable percentage of them subsequently require enucleation for tumor regrowth or uncontrolled neovascular glaucoma. The rate of complications expected after plaques cannot be assessed a priori, because it depends on many radiotherapy-specific, as well as tumor-related, factors. Adverse radiation effects are often delayed and some complications have been shown to increase over time. In the present study, BT alone or combined with other options was the adopted treatment in 49 patients (90.7%). Local surgical resection was performed only in bulbar conjunctival melanomas and was followed by radiotherapy. The 5-year Kaplan–Meier probability of eye-globe preservation was 90.7% and the enucleation rate accords with the reported values (1–24%) [25, 27, 31, 34–36, 47–49].

The LC rate of 96.3% evidenced in the present study was comparable to that in the literature (94–83%) and a direct comparison of these results with other studies should take account of differences in patient selection and tumor characteristics. Twenty-five patients of the present series experienced mainly mild or moderate late adverse events. Enucleation rate due to post-treatment complications was of 3.7%, cataract occurred in 18.6% and optic neuropathy in 3.7% of cases. The use of TTT concurred to consolidate the LC, as well as adjuvant therapy such as intravitreal bevacizumab administration allowed the treatment of radiation retinopathy.The multimodal flexible approach allowed a customized dose-delivering to the target tissue, in order to avoid when possible an excess of radiation dose to healthy structures. In order to treat radiation retinopathy and to reduce postoperative exudative retinal detachment, intravitreal bevacizumab was administered postoperatively. Bevacizumab is a full-length recombinant humanized monoclonal antibody directed against VEGF, originally indicated for the treatment of metastatic colorectal cancer. The antiangiogenic properties of bevacizumab, administered via intravenous infusion or intravitreal injection, have been studied in patients with age-related macular degeneration, choroidal neovascularization, and other retinochoroidal vascular diseases [50–56]. The rationale of the bevacizumab use in the present series was to reduce retinal ischemia that leads to radiation retinopathy and exudative retinal detachment in order to improve visual acuity and to reduce development of neovascular glaucoma. Therefore, the overall radiation retinopathy occurred in 9.2% of cases and the exudative retinal detachment was 11.2%, while the radiation retinopathy incidence reported in literature ranges between 6 and 30%. It should be noted that most adverse events resolved permanently after conservative treatment.

Although the observation time is not very long, the achieved results are encouraging and may be due to a careful pre-planning of the treatment, based on the dose evaluation to be delivered to the structures at risks. Statistical analysis of results obtained in this study shows that the incidence of cases with unfavorable prognosis is significantly different for each one of the three different COMS classes of melanomas. These differences may indicate that genetic anomalies could occur with different frequency in each subgroup. In effect, recent investigations on the incidence of chromosome 3 monosomy in bioptic specimens of UMM [1, 57] evidenced that radiotherapy treatment could fail to achieve disease control more frequently in presence of monosomy 3. Ehlers and co-workers [58], analyzing patterns of aneuploidy in uveal melanoma by comparative genomic hybridization and gene expression profiling, concluded that a relationship between aneuploidy and poor prognosis of large lesions may be determined by specific, pathogenetically relevant chromosomal alterations, rather than overall aneuploidy.

In conclusion, the present study seems to indicate that a multimodal approach is effective in the treatment of uveal melanoma and can reduce the incidence of late adverse effects to an acceptably low rate. Specific genetic alterations can be identified at diagnosis using integrative genomic methods and may suggest novel therapeutic approaches.

Conflict of interest statement The authors declare that they have no conflict of interest to the publication of this article.

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